

Combining Numerical Weather Predictions and Satellite Data for PV Energy Nowcasting

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Abstract—The increasing presence of photovoltaic (PV) generation in the energy mix demands improved forecasting tools which can be updated in an almost continuous basis. Satellite-based information lends itself naturally to this purpose and here it is used to nowcast hourly PV energy production for horizons up to six hours over Peninsular Spain and two islands, Majorca in the Mediterranean Sea and Tenerife in the Atlantic Ocean. This paper compares a single model-based on same-day numerical weather prediction (NWP) with hourly refreshed models which either only use satellite-based measurements or which combine both NWP same-day forecasts and satellite data. As shown in the experiments, the satellite–NWP combination gives very good nowcasting results, clearly superior to those achievable separately by either approach.

Index Terms—Photovoltaic energy, nowcasting, meteosat, support vector regression, numerical weather prediction (NWP).

I. INTRODUCTION

AS REPORTED, for instance, in the highlights on recent developments in renewable power generation that appear in the 2019 Renewable Capacity Statistics compiled by the International Renewable Energy Agency (IRENA) [1], solar energy, particularly that generated by photovoltaics (PV), is probably the fastest growing renewable energy. The large increase in PV installation goes in parallel to a fast-growing presence of solar power in energy markets and already large economic impacts, as shown in the case of Spain in the recent study [2]. While initiatives to reduce greenhouse emissions are urgently needed, it is also likely that they will make more difficult the operation of the electrical grid. As a consequence, it is recognized that improving the forecasting of PV energy will have a prominent role among the tools needed to cope with these situations, particularly in the hourly horizons between one and six hours.

This work concerns itself with this scenario, in which nowcasting capabilities, understood here as frequently updated forecast for horizons up to six hours, would be of considerable help. While numerical weather prediction (NWP) is the cornerstone

of medium-range renewable energy forecasting, its usefulness is much greater for horizons from 6 to 12 hours after the model's analysis time. The main reason for this is that NWP's large computational costs imply that major new forecasts are usually possible only every 12 hours, with 6 hour intermediate updates (although this may be considerably reduced by new systems such as HARMONIE–Arome [3]). Other data sources with a higher refresh frequency are thus needed and, among them, satellite-based measurements are a natural option, provided they are reliably available, have a good quality and lead to improvements over purely NWP-based forecasts.

There is a very large literature addressing many different approaches to solar irradiation and energy forecasting, such as physical, engineering, statistical or machine learning (ML) methods; recent surveys are [4]–[6] and a good comprehensive reference for many aspects of solar energy is [7]. A very well-known example of a physical model is the HELIOSAT method [8] which predicts cloud behavior and irradiances using the measurements of the high resolution, visible channel which are then used to forecast PV energy. On the other hand, while ML-based forecasting has received a large attention, only relatively few papers have used satellite measurements as direct inputs to ML models that predict PV energy. In fact, most of the papers using satellite data focus on building local models to estimate some measure of solar irradiance or cloudiness after which PV energy is predicted. Relevant examples are [9], which proposes support vector regression (SVR) models to predict PV energy for several regions in Germany, [10], which uses artificial neural networks (ANNs) to estimate global solar radiation, or [11], where the authors use historic satellite data collected from the Climate Monitoring SAF to predict short-term PV energy. It is also worth pointing out [12], which presents a model to obtain solar irradiance values from satellite images (although not to directly forecast PV energy values) and to [13], which adopts a probabilistic approach to solar power forecasting.

Other PV-related papers that use SVRs, the main ML tool here, are [14], where SVR models coupled with a PCA-based features are built to predict regional PV energy in Japan, [15], where SVRs are applied over atmospheric motion vectors derived from satellite data to predict several PV energy related variables, [16] and [17], which use SVRs for one-day ahead PV energy predictions from NWP data, and [18], where short term (less than an hour) PV energy forecasts are derived from local data. Also of interest are [19], where the authors use a hybrid SVR model to predict Global Horizontal Irradiance (GHI) monthly values,

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or [20], which focuses on predicting a set of expected solar power production values for future time intervals using SVRs.

As it can be seen from this discussion, PV energy forecasting is a very active research area in which ML models are steadily gaining ground. Input features strongly depend on both the forecasting horizons being considered and the spatial scale of the PV installations. For the 1 to 6 hours intra-day horizons considered here, it can be concluded that satellite based irradiance measures and NWP forecast are the most relevant and widely used input features (perhaps with the addition of past PV productions for the first horizons). While in principle NWP gives forecasts for any hour $H + K$, satellite measures at an hour H have to be extended to estimates for hour $H + K$. At the single farm level the standard for this is to compute a clearness index at hour H and to extend it at hour $H + K$ through Cloud Motion Vector (CMV) estimates [9]; in particular, just a single irradiance measure at the visible channel is used. While natural for individual installations, this is quite more complex when PV forecasts are to be given for a large area; see [25] for such an approach on 921 PV systems in Germany for which individual installed capacity, orientation and tilt are known.

Here, however, we consider PV forecasting on a large area with no individual plant information available. For this situation we propose to combine the atmospheric condition information at hour H directly provided by satellite measures with their future projections at hour $H + K$ given by the NWP forecasts. To this we add clear sky (CS) estimates at hours H and $H + K$ to help bridge the gap between both hours; all these features are considered for each grid point of the area under consideration.

When comparing this with previous work, we point out that we no longer rely on CMV projections to the prediction horizons of satellite irradiance measures at hour H . This certainly has the drawback of having to depend on a static measure at hour H for predictions at hour $H + K$. On the other hand, satellite measure handling is simplified as we skip CMV estimates and, also, we no longer require the physical information needed to compute local clearness indices. Moreover, we can exploit satellite measures at different wavelengths and working with a large area should provide a degree of inertia towards future atmospheric conditions which, when combined with NWP forecasts, would help to better capture atmospheric conditions at hour $H + K$.

Of course, a price to pay for this is large pattern dimensions and high feature correlations, which may largely affect the ML methods to be applied. In previous work [21], [22] we have used Lasso, ANNs and linear and Gaussian SVRs and a simple persistence for 1 to 6 hour intra-day prediction of PV energy production of Peninsular Spain using only grid satellite measures and CS estimates. Notice that the just mentioned high dimension and correlation difficulties were already present. The main conclusion of that work was that Gaussian SVR gave the best results, ANNs were a close but clear second and all other methods were behind. A reason for this lies in the radial structure of the Gaussian kernel, which depends only on pattern distances and not on the weight-feature interplay that takes place in linear models and ANNs. Because of this we have retained Gaussian SVRs here and substantially enlarged the scope of [21], [22] by combining NWP forecasts with satellite and CS grid features and

considering also two more localized (and, hence, demanding) scenarios, namely PV energy prediction on the Majorca and Tenerife islands. In summary, our main contributions are the following:

- We propose to combine direct satellite irradiance measures at different wavelengths, CS estimates and NWP forecasts to compute intra day predictions of aggregated PV production over large areas using Gaussian SVRs.
- We experimentally demonstrate the effectiveness of this proposal on the prediction of the PV energy of Peninsular Spain, and the islands of Majorca and Tenerife, showing that it outperforms forecasts based only on either satellite measures or NWP forecasts, and that its errors decrease gracefully as the horizons increase.

We want to emphasize that, since we do not require any individual plant information, our approach is well suited to a direct use over any region for which satellite measures are available.

The rest of the paper is organized as follows. Section II describe the Meteosat satellite channels used as well as the ECMWF NWP forecasts and also the theoretical CS `pvl` model. Gaussian SVR models are briefly described in Section III, which also gives the details of the concrete models used here. Extensive experiments for Peninsular Spain, Majorca and Tenerife are reported in Sections IV, V and VI, respectively. Finally, Section VII gives a summary of the results and also offers its main conclusions as well as pointers to further work.

II. SATELLITE AND NWP DATA FOR PV ENERGY FORECASTING

We will consider radiation data coming from two different sources, the Meteosat satellites operated by the European Organisation for the Exploitation of Meteorological Satellites (EUMETSAT), and NWP variables provided by the ECMWF. Added to these are clear sky (CS) radiation estimates derived using the model introduced by [26] as implemented in the Python library `pvl` [27]. They are briefly described next.

A. EUMETSAT Satellite Data

Currently, EUMETSAT operates the Meteosat's Second Generation system, whose satellites are equipped with the Spinning Enhanced Visible and Infrared Imager (SEVIRI) technology. Its channel wavelengths range from $0.6 \mu\text{m}$ for the visible Channel 1 to $13.4 \mu\text{m}$ for the long infrared Channel 11. The channels' spatial resolution is about $3 \times 3 \text{ km}$ for most of the covered regions; a finer $1 \times 1 \text{ km}$ resolution is available for the visible channels at wavelengths $0.6\text{--}0.9 \mu\text{m}$ over most of Europe and parts of Africa. A more detailed description of the processing and calibration of the SEVIRI pixel counts is in [28]. The end result of SEVIRI's processing is a total of 22 variables, 11 channel radiances, 3 reflectances (one for each visible wavelength) and 8 brightness temperatures (one for each infrared channel). A Cloud Mask ranging from 0 (clear sky) to 5 (completely covered sky) is also provided, although we will not use it.

In any case, it is clear that pattern dimension can become prohibitively large if all Meteosat measures are to be considered.

Moreover, not all radiations are likely to be relevant for PV energy prediction, particularly those of the infrared wavelengths. This was considered in [21], where the correlations between the Peninsular average of each satellite measure and the PV production of Peninsular Spain were computed. It was found that the largest correlations were those of the radiances of the infrared channels IR_016 and IR_039 and the visible channel VIS008, and the brightness temperature of the IR_039 channel. Other variables with high correlation were the IR_016 and VIS008 reflectances, but they had a straight correlation of 1 with their radiance counterparts. As a consequence, and as done in [21], we have selected the brightness temperature of channel IR_039 and the radiances of the IR_016, IR_039 and VIS008 channels. The same satellite variables are considered for Majorca and Tenerife.

B. ECMWF NWP Forecasts

The NWP variables to be considered are hourly forecasts of 1) temperature at 2 meters (t_{2m}), 2) surface solar radiation downwards ($ssrd$), 3) surface net solar radiation (ssr), 4) total cloud cover (tcc), and 5) wind speed module at 10 meters (v_{10}). We retrieved their values from the ECMWF's MARS repository for a 0.125° resolution rectangular grid over Peninsular Spain and the Majorca and Tenerife islands.

Notice that the ECMWF offers forecast for quite a few more radiation-related variables. On the other hand, the radiation variable more often considered for solar energy prediction is the global horizontal irradiance (GHI) which corresponds to the ECMWF's $ssrd$. See [29] for a comparison of numerical weather prediction sources for solar irradiance forecasts.

In principle, NWP variables are not well suited for nowcasting purposes, as they are not updated in a continuous basis; for instance, the ECMWF runs two main forecasts starting at 00 and 12 UTC that become available at about 05 and 17 UTC each day. There are also two updates starting at 06 and 18 UTC, also available at about 11 and 23 UTC. Notice that for Spain this means that new ECMWF forecasts are available essentially at sunrise and, thus, are likely to be competitive with any nowcasting alternative at least until the early afternoon. Similarly, the 06 UTC run update could be used for afternoon nowcasts. On the other hand, the NWP forecasts of the 12 UTC run would apply to the late evening and night hours, having thus much less interest (and more so for the 18 UTC update). Here we will only use the ECMWF's 00 UTC forecasts.

C. `pvlib` Clear Sky Model

The ECMWF give predictions for a number of clear sky-related variables, such as the surface net solar radiation, clear sky ($ssrc$) and the clear sky direct solar radiation at surface ($cdir$). However, it is simpler to work with physical clear sky models; [30] is a review of the very active research in physical modeling of solar radiation for renewable energy.

We will use the CS model introduced by [26] as implemented in the Python library `pvlib` [27] and compute its hourly values at the points of the ECMWF's 0.125° grid. Several parameters can be incorporated into the model but we have left them at their

`pvlib` defaults, except for the altitude at each grid point which we obtain from its geopotential in the ECMWF's grid. We add the values of these CS GHI estimates at hours H and $H + K$ as features when predicting PV energy at an hour $H + K$ from satellite readings at hour H .

D. Data

We work with hourly PV energy production kindly provided by Red Elctrica de Espaa (REE). For an UTC hour H , the y_H energy value corresponds to the PV energy produced during the entire hour ending at H . For the Peninsula PV energy is normalized to a 0–100 range; in Majorca and Tenerife they are given in MWh. Majorca's installed power is 72.5 MW and Tenerife's 107.8MW. Only daylight hours are considered, dropping data values outside the UTC 5–20 range for the Peninsula and Majorca, and 6–21 for Tenerife. After this preprocessing, at most 4,645 hourly values per year are left; actual values may be slightly smaller for some years due to some missing data.

For the ECMWF variables for the Peninsula a rectangular area is first considered with a northwest corner at $(44.0^\circ, -9.5^\circ)$ latitude-longitude and a southeast corner at $(35.5^\circ, 4.5^\circ)$; this gives a total of 7,797 points at the 0.125° resolution, which is further reduced so that the grid points considered fall into the peninsular area; this results in a final number of 3,391 grid points. For Majorca and Tenerife rectangular grids are considered with northwest corners at $(41.0^\circ, 1.0^\circ)$ and $(30.5^\circ, -19.0^\circ)$ respectively, and southeast corners at $(38.0^\circ, 4.5^\circ)$ and $(25.5^\circ, -14.0^\circ)$ respectively. The number of grid points for Majorca is 675 and 1,681 for Tenerife. The dimensions over the five ECMWF features are thus $16,955 = 3,391 \times 5$ for the Peninsula, $3,375 = 675 \times 5$ for Majorca and $8,405 = 1,681 \times 5$ for Tenerife.

Finally, 15-minute satellite readings have been downloaded from <http://www.eumetsat.int/website/home/Data/DataDelivery/EUMETSATDataCentre/index.html>

EUMETSAT's data centre for the same 0.125° grid points of the ECMWF data. In order to bring them to hourly values, the four 15-minute Meteosat satellite variable values available up to a given hour H are averaged. The dimensions of the hourly satellite variables are thus 13,564 for the Peninsula, 2,700 for Majorca and 6,724 for Tenerife.

III. SUPPORT VECTOR REGRESSION MODELS

A. Gaussian Support Vector Regression

As mentioned above, in previous work [21] several well-established ML models, namely Lasso and linear and Gaussian kernel Support Vector Regression (SVR), have been considered for purely satellite-based PV nowcasting over Peninsular Spain. A simple persistence model $\hat{y}_{H+K} = \frac{CS_{H+K}}{CS_H} y_H$ with y_H and $\hat{y}_H$ actual and estimated energy and CS_H clear sky values, was also evaluated. The main conclusion from [21] is that Gaussian SVRs clearly provide the best results. On the other hand, simpler linear models such as Lasso or linear SVR were clearly behind. Because of this, only Gaussian SVR are considered here; they are briefly described next.

Given a sample $S = \{(\mathbf{x}^{(1)}, y^{(1)}), \dots, (\mathbf{x}^{(N)}, y^{(N)})\}$ with N patterns, linear Support Vector Regression (SVR) tries to adjust an optimal model $\mathbf{w}^* \cdot \mathbf{x} + b^*$ to the sample S by solving the problem of minimizing

$$f(\mathbf{w}, b, \xi, \eta) = \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{p=1}^N (\xi_p + \eta_p) \quad (1)$$

with the following constraints on the model errors $\mathbf{w} \cdot \mathbf{x}^{(p)} + b - y^{(p)}$:

$$\begin{aligned} -\xi_p - \epsilon &\leq \mathbf{w} \cdot \mathbf{x}^{(p)} + b - y^{(p)}; \\ \eta_p + \epsilon &\geq \mathbf{w} \cdot \mathbf{x}^{(p)} + b - y^{(p)}; \\ \xi_p, \eta_p &\geq 0. \end{aligned}$$

In practice (see [31]), one minimizes a simpler dual problem, derived from (1) via Lagrangian theory, namely

$$\begin{aligned} \Theta(\alpha, \beta) = \frac{1}{2} \sum_{1 \leq p, q \leq N} (\alpha_p - \beta_p)(\alpha_q - \beta_q) \mathbf{x}^{(p)} \cdot \mathbf{x}^{(q)} \\ + \epsilon \sum_{p=1}^N (\alpha_p + \beta_p) - \sum_{p=1}^N y^{(p)} (\alpha_p - \beta_p) \end{aligned} \quad (2)$$

subject to the constraints

$$\begin{aligned} 0 &\leq \alpha_p, \beta_q \leq C; \\ \sum_{p=1}^N \alpha_p &= \sum_{q=1}^N \beta_q. \end{aligned}$$

Here α and β are the Lagrange multipliers associated to the so-called primal problem (1). Once the optimal multipliers $\alpha^* = (\alpha_1^*, \dots, \alpha_N^*)$, $\beta^* = (\beta_1^*, \dots, \beta_N^*)$ are found, the optimal weight $\mathbf{w}^* = \sum (\alpha_p^* - \beta_p^*) \mathbf{x}^{(p)}$ and bias b^* of the SVR model are obtained from them through the Karush–Kuhn–Tucker (KKT) optimality conditions (see [31] for more details).

It turns out that (2) only involves dot products $\mathbf{x}^{(p)} \cdot \mathbf{x}^{(q)}$; moreover, the same is true for the Sequential Minimization Algorithm (SMO; [32]) used to solve (2) and also for the final model, which can be expressed as

$$f(\mathbf{x}) = b^* + \mathbf{w}^* \cdot \mathbf{x} = b^* + \sum (\alpha_p^* - \beta_p^*) \mathbf{x}^{(p)} \cdot \mathbf{x}. \quad (3)$$

The kernel trick takes advantage of these facts to replace the initial dot products $\mathbf{x} \cdot \mathbf{x}'$ with the values $k(\mathbf{x}, \mathbf{x}')$ of a positive definite kernel k that can be written as $k(\mathbf{x}, \mathbf{x}') = \phi(\mathbf{x}) \cdot \phi(\mathbf{x}')$, where the $\mathbf{x}$ are mapped through $\phi(\mathbf{x})$ into a larger, possibly infinite, dimensional Hilbert space. Notice that once k is selected, the mapping ϕ is no longer needed to write (2) or to solve it. The most used kernel is the Gaussian one, $k(\mathbf{x}, \mathbf{x}') = e^{-\gamma \|\mathbf{x} - \mathbf{x}'\|^2}$, which leads to a final model of the form

$$\begin{aligned} f(\mathbf{x}) &= b^* + \sum (\alpha_p^* - \beta_p^*) \phi(\mathbf{x}^{(p)}) \cdot \phi(\mathbf{x}) \\ &= b^* + \sum (\alpha_p^* - \beta_p^*) e^{-\gamma \|\mathbf{x}^{(p)} - \mathbf{x}\|^2}. \end{aligned} \quad (4)$$

An optimal choice of the three hyper-parameters C , ϵ and γ is crucial in order to build an effective SVR model. In particular, we

will scale input patterns to a $[0, 1]$ range in order to facilitate the estimate of the optimal γ . We give more details on this in the next Subsection III-B. In the experiments below the *Scikit-learn*'s wrapper [33] of the very well-known LIBSVM [32] library will be used.

B. Models Considered

Recall that Gaussian SVRs will be used in all cases. In the reference model, hourly PV energy productions for a day D are predicted by a single SVR model from the same-day ECMWF NWP hourly forecasts produced by the 00 UTC model. This model will be called the D (same-day) one.

Nowcasting models for horizons $K = 1, \dots, 6$ hours ahead will be also considered. They will rely either only on satellite-based measures plus CS estimates or on their combination with NWP forecasts. The first ones will be referred to as S_K , and as C_K the second ones, with $1 \leq K \leq 6$.

The S_K models will predict PV energy at an hour H between $05 + K$ and 20 UTC for the Peninsula and Majorca and $06 + K$ to 21 UTC for Tenerife, using satellite-based readings at hour $H - K$ plus CS estimates at hours $H - K$ and H . Obviously, such a model stops predicting after hour $20 - K$ for the Peninsula and Majorca and after hour $21 - K$ for Tenerife. Since four satellite-based measures plus two CS values will be used, pattern dimension for the S_K models will be $20, 346 = 3, 391 \times 6$ for the Peninsula, $4, 050 = 675 \times 6$ for Majorca and $10, 086 = 1, 681 \times 6$ for Tenerife.

On the other hand, the combined NWP–satellite-based models C_K , $1 \leq K \leq 6$, follow the same scheme of the S_K models, predicting PV energy at an hour H of a given day D using as features at each grid point four satellite-based readings at hour $H - K$, CS estimates for hours $H - K$ and H and five NWP forecasts for hour H generated in the 00 UTC pass of day D of the ECMWF model. Since the total number of variables per grid point is now 11, pattern dimension for the C_K models will be almost twice that for the S_K models, namely $37,301$ for the Peninsula, $7,425$ for Majorca and $18,491$ for Tenerife. Finally, the base D model predicts PV energy at hour H from the NWP variables for the same hour; pattern dimension is $16,955$ for the Peninsula, $3,375$ for Majorca and $8,405$ for Tenerife.

Thus, 13 different SVR models have to be considered for each geographic scenario, and for each one optimal C , ϵ and γ hyperparameters need to be estimated. This is done using the `GridSearchCV` class in *Scikit-learn* to explore the following logarithmically equispaced grid:

- $C \in \{4^k : -5 \leq k \leq 6\}$.
- $\epsilon \in \{4^{-k} \sigma : 1 \leq k \leq 6\}$ where σ denotes the standard deviation of the PV target y over the training year. Notice that σ would be the mean square error of the constant model $\hat{y} = \bar{y}$ (σ values are 26.31% for Peninsular PV, 14.19 MWh for Majorca and 29.87 MWh for Tenerife). Our largest ϵ value is thus $\sigma/4$, as the models are expected to have an error smaller than that.
- $\gamma \in \{4^{\frac{k}{d}} : -2 \leq k \leq 3\}$ where d denotes the number of features for each problem. Prior to this the initial patterns will be linearly scaled into the $[0, 1]$ range; notice that then

TABLE I
HOURLY AND DAILY AVERAGE TEST ERRORS FOR THE D , S_K AND C_K
FOR PENINSULAR SPAIN

Model	K	Hour													Av.
		8	9	10	11	12	13	14	15	16	17	18	19	20	
D		2.19	3.15	3.76	3.77	3.66	3.53	3.53	3.36	2.95	2.18	1.27	0.73	0.56	2.67
S_K	1	1.56	2.38	2.67	2.55	2.53	2.37	2.36	2.23	2.19	1.67	1.02	0.43	0.28	1.87
	2	1.85	3.04	3.46	2.90	2.76	2.62	2.52	2.48	2.30	1.88	1.25	0.61	0.36	2.16
	3	2.20	3.58	4.46	3.99	3.34	3.09	2.85	2.80	2.58	1.88	1.21	0.59	0.35	2.53
	4	-	3.50	4.25	4.30	4.12	3.70	3.45	3.17	3.02	2.25	1.54	1.11	1.03	2.95
	5	-	-	4.40	4.53	4.63	4.37	4.09	3.83	3.38	2.53	1.46	0.95	0.78	3.18
	6	-	-	-	5.19	5.19	5.09	4.60	4.15	3.89	2.75	1.46	0.69	0.56	3.36
C_K	1	1.47	2.18	2.52	2.39	2.46	2.39	2.30	2.13	1.91	1.53	1.06	0.60	0.51	1.81
	2	1.55	2.35	2.73	2.65	2.63	2.63	2.46	2.38	2.16	1.57	1.07	0.67	0.57	1.95
	3	1.68	2.55	2.97	2.91	2.86	2.80	2.74	2.56	2.36	1.70	1.18	0.72	0.61	2.13
	4	-	2.82	2.99	2.89	2.91	2.95	2.80	2.59	2.40	1.77	1.08	0.67	0.55	2.20
	5	-	-	3.18	2.90	2.93	2.99	2.96	2.67	2.42	1.82	1.06	0.64	0.55	2.19
	6	-	-	-	3.05	2.94	3.01	3.06	2.80	2.43	1.81	1.08	0.60	0.50	2.13

$\|\mathbf{x}^{(p)} - \mathbf{x}\|^2 \leq d$, with d the pattern dimension. This is why we consider Gaussian dilations of the form $\gamma = 4^k/d$.

Given the temporal ordering of PV production, the year 2013 will be used as the training set and 2014 as the validation set. After the best hyper-parameters are found, the corresponding models are built on both the training and validation sets and then their predictions over the 2015 test set are computed. We point out that, usually, the most relevant hyperparameters are C and γ as the models best performing in CV have common C and γ values with different but also close ϵ values. We observe that SVR training times depend on sample size and on the concrete hyperparameters used, with C being often the most influential. For space reasons we cannot report training times for all 39 models but taking as an example the C_2 models, the optimal C hyperparameter was 64 for the Peninsula, 256 for Majorca and 1024 for Tenerife. Accordingly, training times were about 30' for the Peninsula, 40' for Majorca and one hour for Tenerife, all in a medium power blade server.

Next, the D , S_K and C_K models, $1 \leq K \leq 6$, will be compared separately for Peninsular Spain, Majorca and Tenerife. For a more homogeneous comparison errors we report within the time range between 08 and 20 UTC hours for $K = 1, 2, 3$, while the hour range will start at 09 UTC for $K = 4$, at 10 UTC for $K = 5$ and at 11 UTC for $K = 6$.

IV. RESULTS FOR PENINSULAR SPAIN

A summary of the overall hourly average absolute test errors for the year 2015 when predicting Peninsular PV production is given in Table I for each model and prediction horizon; the rightmost column gives the daily average of these hourly errors. Recall that there are 13 models in total: the single NWP D model and 6 S_K and C_K models.

Looking first at the daily averages, those of the C_K models are noticeably smaller than those of both the corresponding S_K models and those of the D model (the average errors of models C_4 , C_5 and C_6 appear as decreasing but notice that, for instance, the C_6 average does not include the difficult hours 9 and 10). Furthermore, the daily error averages of the S_K models are smaller than those of the D model when $K = 1, 2, 3$, but larger afterwards.

Considering hourly errors, it can be seen that for a given hour H the S_K and C_K errors increase with K (with a few exceptions

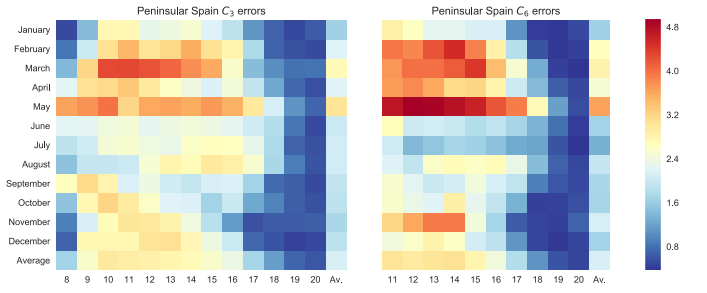


Fig. 1. Gaussian SVR test errors of C_3 and C_6 models per hour and month for Peninsular Spain.

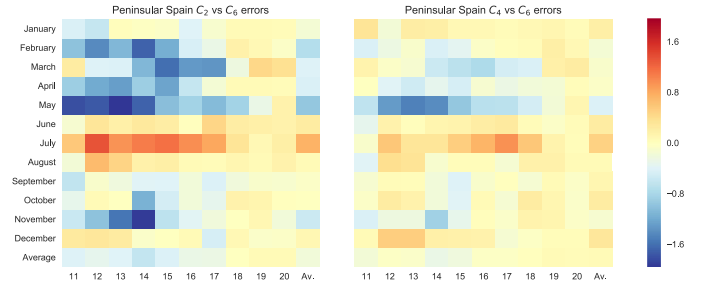
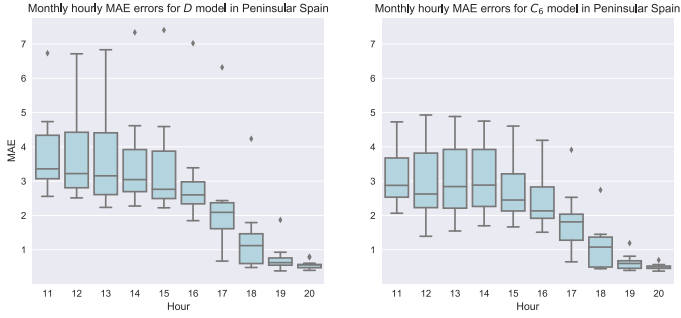


Fig. 2. Differences between the monthly and hourly error averages of Peninsular Spain models C_2 and C_6 (left) and C_4 and C_6 (right). Blue colors indicate lower C_2 and C_4 errors; red ones the opposite.

at the evening hours). Moreover, the hourly errors of the C_K models are also smaller than those of the S_K models and also of these of the D models. In order to give a more objective comparison, a pair-wise Wilcoxon signed-rank test [34] has been applied, comparing at a given horizon K the C_K , S_K and D models, using as the samples all their hourly absolute errors for the entire 2015 test set, and applying a Bonferroni correction for multiple comparisons. Its results at the 5% level show that the C_K models give smaller errors for all horizons, that the S_K models are better than the D model when $K = 1, 2$ but worse for $K = 4, 5, 6$, and that the S_3 and D models give similar errors. This confirms the better performance of the C_K models; we concentrate on them from now on.

We analyze the behavior of the C_K models along the different months of the 2015 test year in Fig. 1, where monthly error averages are given as heatmaps for all hours considered in the $K = 3$ and 6 horizons. As it can be seen, hourly errors appear quite balanced on all months (although, of course, PV production is lower on the months from November to March and relative errors would therefore be higher). Also, maximum errors typically appear in noon hours, where production is usually highest. It is also observed that in a few months the C_6 model performs slightly better than the C_2 and C_4 . This can be seen in Fig. 2 which depicts for each month and each hour from 11 to 20 UTC the differences between the errors of the C_2 and C_6 models (left) and C_4 and C_6 models (right); red colors imply that C_6 errors are lower than the C_2 and C_4 ones. It can be observed that the performance of model C_6 is better in July and, although less markedly so, also in June and August (model C_2 also performs slightly worse than model C_4 in these months).

Fig. 3. D and C_6 hourly error boxplots for 2015 and Peninsular Spain.TABLE II
HOURLY AND DAILY AVERAGE TEST ERRORS FOR THE D , S_K AND C_K FOR THE ISLAND OF MAJORCA

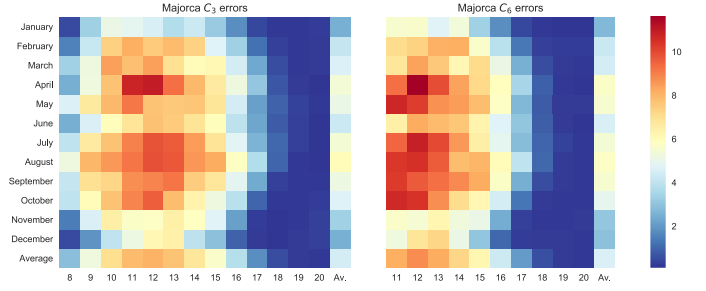
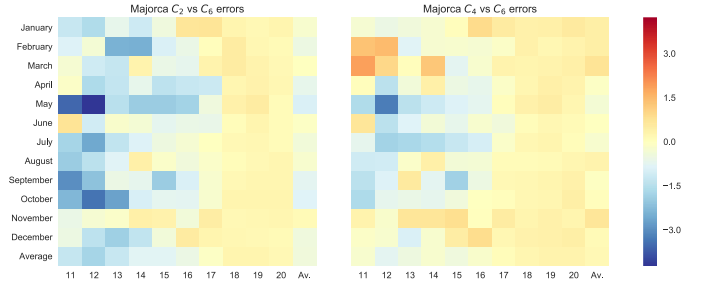
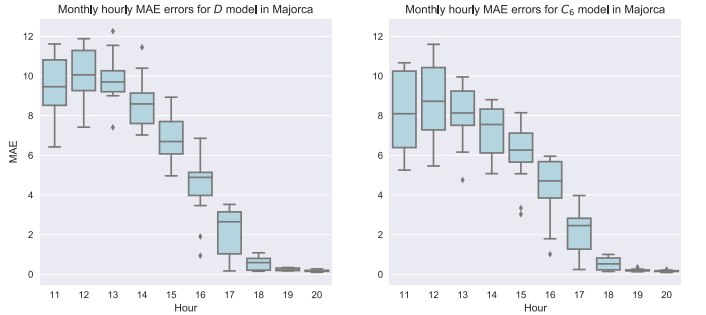
Model	K	Hour														Av.
		8	9	10	11	12	13	14	15	16	17	18	19	20		
S_K	D	3.22	5.51	7.67	9.46	10.06	9.81	8.59	6.85	4.50	2.15	0.59	0.24	0.17	5.29	
	1	2.51	3.91	5.14	6.18	6.86	7.23	6.44	5.12	3.46	1.77	0.54	0.16	0.04	3.80	
	2	2.36	4.90	5.84	6.73	7.79	7.72	7.08	5.95	3.98	2.02	0.56	0.17	0.03	4.24	
	3	2.64	5.36	6.81	7.37	8.19	8.41	7.56	6.36	4.41	2.21	0.74	0.23	0.05	4.64	
	4	–	5.36	8.14	8.73	8.73	8.42	8.02	6.74	4.63	2.30	0.63	0.19	0.03	5.16	
	5	–	–	8.03	9.40	9.47	9.03	8.13	6.99	4.90	2.49	0.62	0.23	0.04	5.39	
	6	–	–	–	9.38	10.08	9.78	8.87	7.35	5.07	2.56	0.62	0.24	0.08	5.40	
C_K	1	2.45	4.53	5.65	6.01	6.34	6.55	5.81	4.69	3.40	1.78	0.81	0.44	0.30	3.75	
	2	2.69	5.03	6.31	6.82	6.96	6.92	6.43	5.40	3.75	1.96	0.81	0.47	0.35	4.15	
	3	2.84	5.25	6.98	7.82	8.05	7.64	6.70	5.65	3.79	1.79	0.50	0.21	0.18	4.42	
	4	–	5.42	7.16	7.88	7.97	7.64	7.00	5.71	4.10	2.13	0.81	0.52	0.48	4.74	
	5	–	–	6.97	8.38	8.70	8.21	7.16	6.25	4.20	2.00	0.54	0.23	0.17	4.80	
	6	–	–	–	8.10	8.72	8.14	7.20	6.17	4.28	2.04	0.55	0.20	0.16	4.56	

Finally, it is also of interest to consider the error spread of these models. Fig. 3 shows the boxplots of the 2015 hourly errors of the D and C_6 models for Peninsular Spain; it illustrates clearly the advantage of the C_6 model. Note that only hours present in both models are shown in the boxplots.

V. RESULTS FOR MAJORCA

The presentation here is organized as in the previous section for Peninsular Spain. First, Table II summarizes hourly average absolute test errors now in MWh for 2015 for each model and prediction horizon; the rightmost column gives the daily average of these hourly errors. Looking at the daily averages for the 13 models, the C_K models for Majorca also have clearly smaller errors than those of the corresponding S_K and D models. Here the hourly error averages of the S_K models are smaller than those of the D model until at least UTC hour 13 when $K = 6$, but they become larger afterwards. In fact, applying as before a Wilcoxon signed-rank test, its results at the 5% level show again that the C_K models give smaller errors for all horizons, with a tie between the S_2 and C_2 models, and that the S_K models give smaller errors than the D models for the horizons $K = 1, 2, 3$ and 4; they tie for $K = 5$ and the D model beats S_6 . A reason for this better behavior of the S_K models may be the difficulty for a purely NWP-based model to give accurate PV energy forecasts at the reduced scale of the island of Majorca, in contrast with the much smoother production of Peninsular Spain.

Concentrating now on the C_K models, the absolute errors of the C_3 and C_6 models for Majorca in MWh by month and hour are shown as heatmaps in Fig. 4. They are relatively similar,

Fig. 4. Gaussian SVR test errors of C_3 and C_6 models per hour and month for Majorca.Fig. 5. Differences between the monthly and hourly error averages of Majorca models C_2 and C_6 (left) and C_4 and C_6 (right). Again, blue colors indicate lower C_2 and C_4 errors; red ones the opposite.Fig. 6. D and C_6 hourly error boxplots for 2015 and Majorca.

although with a natural advantage for the C_3 model. Similarly, Fig. 5 shows in a heatmap the differences between the month-hour errors of the C_2 and C_4 models against the C_6 ones. As it can be seen, the hourly C_2 errors are smaller (blue colors in Fig. 5) than the C_6 ones in almost all months up to 17 UTC, with a small advantage for the C_6 errors afterward; moreover, the C_2 monthly averages are also smaller on almost all months. In contrast, the C_4 and C_6 have a much more similar performance, with the monthly averages of the C_6 models being slightly smaller. In a comparison of the C_2 and C_4 errors (not shown), the former are smaller for most of the month-hour combinations.

To finish this section, the boxplots of the 2015 hourly errors of the D and C_6 models for Majorca are shown in Fig. 6. Again, they clearly show the superior performance of the C_6 model. Notice that errors for Majorca are given in MWh. To compare them with those of the Peninsula, they should be divided by the installed power. If done for instance for the C_6 model, its error

TABLE III
HOURLY AND DAILY AVERAGE TEST ERRORS FOR THE D , S_K AND C_K
FOR THE ISLAND OF TENERIFE

Model	K	Hour															Av.
		8	9	10	11	12	13	14	15	16	17	18	19	20			
D		1.28	3.70	5.89	7.53	8.69	10.29	10.93	10.05	9.05	6.68	4.22	1.49	0.72	6.19		
S_K	1	0.96	2.93	4.20	4.98	6.28	6.96	6.62	5.76	5.64	4.22	2.82	1.24	0.40	4.08		
	2	0.97	3.22	4.95	6.39	7.65	9.30	8.96	7.84	7.23	4.98	3.25	1.28	0.39	5.11		
	3	1.01	3.48	5.50	7.07	8.79	10.41	10.60	9.36	8.20	6.01	3.58	1.38	0.49	5.84		
	4	–	3.66	5.91	7.81	10.23	12.01	12.00	11.35	9.40	6.50	3.87	1.34	0.51	7.05		
	5	–	–	5.83	7.53	10.18	12.58	12.52	11.68	9.97	7.13	4.10	1.55	0.77	7.62		
	6	–	–	–	8.27	10.36	13.38	13.89	13.11	11.04	7.86	4.32	1.43	0.64	8.43		
C_K	1	1.23	3.53	4.39	5.64	6.50	7.68	7.53	6.62	6.10	4.67	2.90	1.73	0.63	4.55		
	2	1.33	3.50	4.68	6.39	7.34	8.73	8.90	7.54	6.72	5.12	3.39	1.76	0.84	5.09		
	3	1.12	3.49	5.07	6.81	7.81	9.42	9.92	8.86	7.63	5.58	3.47	1.39	0.71	5.48		
	4	–	3.56	5.23	6.96	8.27	9.81	10.23	9.40	8.16	5.91	3.70	1.46	0.85	6.13		
	5	–	–	5.54	7.02	8.42	10.20	10.56	9.64	8.40	6.18	3.88	1.40	0.89	6.56		
	6	–	–	–	7.16	8.43	10.54	11.13	10.05	8.71	6.03	3.80	1.67	1.25	6.88		

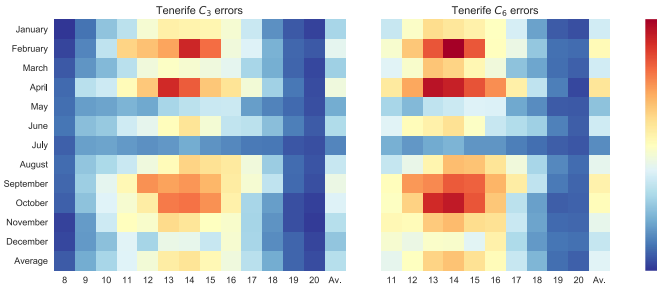


Fig. 7. Gaussian SVR test errors of C_3 and C_6 models per hour and month for Tenerife.

in percentage is 6.29%, about three times higher than that of the Peninsula's C_6 model.

VI. RESULTS FOR TENERIFE

Finally, in this Section the same presentation is applied to the Tenerife results. Table III summarizes hourly average absolute test errors in MWh for 2015 for each model and prediction horizon and its rightmost column gives the daily average of these hourly errors. Looking for simplicity at the daily averages, here again the C_K models outperform the others; S_K models beat the D one for the $K = 1, 2$ horizons, their performance is similar for $K = 3, 4$ and D clearly beats the S_K for the $K = 5, 6$ horizons. These observations are confirmed by a Wilcoxon signed-rank test comparison of the models C_K , S_K and D for each horizon K ; its results at the 5% level show again that the C_K models give smaller errors for all horizons, that the S_K models are better than the D model when $K = 1, 2$ but worse for $K = 4, 5, 6$, and that the S_3 and D models essentially tie.

When looking at the performance by month and hour of Tenerife's C_3 and C_6 models, the absolute error values in MWh of the heatmap in Fig. 7 show a situation similar to that of Majorca. It can also be seen in the heatmaps of Fig. 8 that the C_2 and C_4 models have slightly smaller errors than the C_6 model for almost all months and hours. As it was the case in Majorca and although not shown, here the C_2 errors are also smaller than the C_4 ones for most of the month–hour combinations.

Finally, the boxplots of the 2015 hourly errors of the D and C_6 models for Tenerife are given in Fig. 9, which clearly shows again the superior performance of the C_6 model. Recall that errors for Tenerife are also given in MWh. If given in

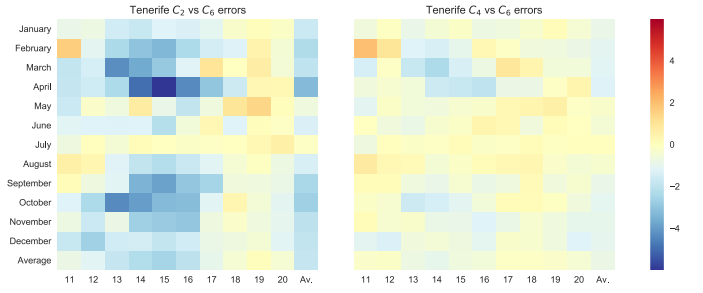


Fig. 8. Differences between the monthly and hourly error averages of Tenerife models C_2 and C_6 (left) and C_4 and C_6 (right).

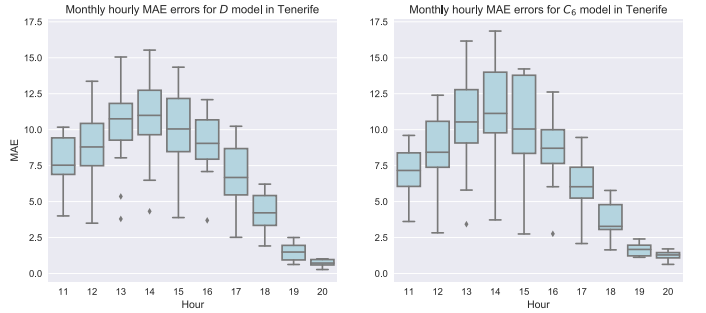


Fig. 9. D and C_6 hourly error boxplots for 2015 and Tenerife.

percentages, the error of, say, the C_6 model would be 6.38%, similar to that of Majorca's and about three times higher than that of the Peninsula.

VII. CONCLUSION AND FURTHER WORK

This work has studied the direct use of satellite-based radiation-related measures for the nowcasting of PV energy production over a large area, that of Peninsular Spain, and also over two more reduced areas, the islands of Majorca on the Mediterranean Sea and Tenerife on the Atlantic Ocean. In contrast to other approaches that convert satellite measures to surface radiation estimates and then nowcast these estimates to derive PV predictions, here a selection of satellite-based readings at an hour $H - K$ is used as features to be directly exploited by ML models to predict PV production at hour H , for horizons $1 \leq K \leq 6$.

Three modeling scenarios have been considered, all of them built using Support Vector Regression. In the first one, called model D , same-day PV predictions are derived from the NWP variables computed in the ECMWF run that starts at 00 UTC. Notice that they are available at about 05 UTC and can thus be used in Spain for PV forecasts on the same day for sunlight hours. The forecasts of model D give a quite competitive benchmark against which other models can be compared. In our second scenario (models named S_K) satellite readings and CS model estimates are the sole features used, now in a nowcasting framework as readings can be updated hourly (or even more frequently, since satellite-based measures are available every 15 min). Finally, the NWP features of model D and the satellite-based ones of the S_K models are combined in the third, and most powerful, C_K models.

The experiments show the C_K models to be clearly the winners for all horizons. Moreover, although prediction errors must obviously increase with K , the performance of the C_K models deteriorates gracefully, something likely to be due to the presence of NWP features, that help stabilize the C_K models for the longer horizons. Summing things up, it has been shown that the combination of good NWP forecasts with hourly readings of satellite data and powerful ML tools result in nowcasting models with a good performance.

In any case, there are issues that deserve further research. A clear one is the very large dimensions of the features used and their very high correlations. SVR models are quite robust with respect to feature correlation but, in any case, large dimensions imply large computation costs (this is particularly the case with SVRs) and it is quite likely that adequately selected reduced features could give models with a performance close to that reported here but with smaller computational costs. This selection could be done by handling each feature independently or by exploiting somehow the natural groupings either between the same variable at different grid points or, instead, considering the groupings of different features when given for the same grid point. Another question of obvious interest is the performance of the proposed models at the individual PV plant scale, a scenario where features derived through Cloud Motion Vectors can be added to the approach here, as it has been shown to be very effective.

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